BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 30 Min. | **Maximum Marks:** 20

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. (0,0) is the solution of inequality :

(A) $3x - 7y < 3$

(B) $x + y > 2$

(C) $x - y > 1$

(D) $3x + 5y > 7$

Q2. The slope of a line with inclination 90 degrees is :

(A) 0

(B) -1

(C) Undefined

(D) 1

Q3. If a and b are parallel vectors then a x b = :

(A) 0 (zero vector)

(B) -1

(C) 1

(D) 2

Q4. Two lines $a1x + b1y + c1 = 0$, $a2x + b2y + c2 = 0$ are parallel if:

(A) $a1/a2 = b1/b2$

(B) $a1/a2 = b1/b2$ [sic, printed identical to option A]

(C) $b1/c1 = b2/c2$

(D) $a1/c1 = a2/c2$

Q5. The value of 3 j, k x i is: [sic, as printed]

(A) -1 (B) -3

(C) 3 (D) 0

Q6. If a straight line is parallel to x-axis, then its slope is :

(A) Undefined (B) -1

(C) 1 (D) 0

Q7. The centre of the circle 5x2 + 5y2 + 24x + 36y + 10 = 0 is:

(A) (12/5, 18/5) (B) (-12/5, -18/5)

(C) (12/5, -18/5) (D) (-12/5, 18/5)

Q8. The length of the latus rectum of the parabola y2 = 8x is:

(A) 2 (B) 8

(C) 4 (D) 2 sqrt(8)

Q9. The point of intersection of angle bisectors of a triangle is called :

(A) Orthocentre (B) Centroid

(C) In-centre (D) Circumcentre

Q10. The coordinates of the vertices of hyperbola y2/16 - x2/49 = 1 are:

(A) (0, +-7) (B) (+-4, 0)

(C) (0, +-4) (D) (+-7, 0)

Q11. d/dx (sin 2x + cos 2x) = :

(A) (cos 2x - sin 2x) (B) (cos 2x + sin 2x)

(C) (2 cos 2x + 2 sin 2x) (D) 2(cos 2x - sin 2x)

Q12. lim (h->0) (1 + 2h)^(1/h) = :

(A) e2 (B) e

(C) 1/e (D) 1/e2

Q13. Integral of e^sinx cosx dx = :

(A) e^cosx + c (B) ln|sinx| + c

(C) ln|cosx| + c (D) e^sinx + c

Q14. d/dx (cot-1 x) = :

(A) [numerator cut in source scan] / (1+x2) (B) [numerator cut in source scan] / (1-x2)

(C) -1/(1+x2) (D) -1/(1-x2)

Q15. Integral of e^x (cos x + sin x) dx = :

(A) e^-x sinx + c
(B) e^x sinx + c
(C) -e^x sinx + c
(D) e^-x cosx + c

Q16. If y = e^-ax, then dy/dx = :

(A) ae^-ax
(B) e^-ax
(C) a2e^-ax
(D) -ae^-ax

Q17. Integral of 1/(1+x2) dx = :

(A) tan-1 x + c
(B) -tan-1 x + c
(C) sin-1 x + c
(D) cos-1 x + c

Q18. The range of f(x) = sqrt(x2 - 9) is:

(A) (-infinity, 0]
(B) [0, +infinity)
(C) (0, +infinity)
(D) (-infinity, infinity) [partially cut in source scan]

Q19. Integral of sinx cosx dx = :

(A) ln|sin x| + c
(B) cos2x/2 + c
(C) sin2x/2 + c
(D) (sin2x cos2x)/2 + c

Q20. d/dx (1/cosec x) = :

(A) 1/sec x
(B) cosec2x
(C) cot x
(D) 1/cosec2x

Essay Type

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Express perimeter "P" of a square as a function of its area "A".

(ii) Find f'(x) for f(x) = 2x + 3

(iii) Evaluate lim (x->0) sin x degrees / x

(iv) Define rational function with example.

(v) Evaluate lim (x->infinity) (x/(1+x))^x

(vi) Find dy/dx from first principle if y = sqrt(x + 2)

(vii) Differentiate w.r.t. x if y = (x2 + 1)/(x2 - 3)

(viii) Find dy/dx if xy + y2 = 2

(ix) Find derivative w.r.t. x if y = cot-1(x/a)

(x) Find dy/dx if y = log10 (ax2 + bx + c)

(xi) Apply the Maclaurin Series to prove that e^(2x) = 1 + 2x + 4x2/[2 + 8x3/[3 + ----- [sic, denominators cut in source scan]

(xii) Define increasing function with example.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Find delta y and dy in y = sqrt(x), when x changes from 4 to 4.41

(ii) Evaluate the integral of (sqrt(theta) - 1)2 / theta d theta, theta > 0

(iii) Find integral of 1/(x(4 ln x)) dx [sic, as printed]

(iv) Evaluate the integral of (x + 2)/sqrt(x + 3) dx

(v) Using by part method to evaluate integral of x2 ln x dx [sic, as printed]

(vi) Evaluate the definite integral from 0 to pi/3 of cos2 theta sin theta d theta

(vii) Find the area between the x-axis and the curve y = cos (1/2) x from x = -pi to pi

(viii) Solve the differential equation sin y cosec x dy/dx = 1

(ix) Find h such that A (-1, h), B (3, 2), C (7, 3) are collinear.

(x) Two points P (-5, -3) and Q (-2, -6) are given in XY-coordinate, find the coordinate of P in xy-coordinate system. [sic, as printed]

(xi) Find equation of the line having x-intercept -3 and y-intercept 4.

(xii) Find the distance from the point P (6, -1) to the line 6x - 4y + 9 = 0

Q4. (Marks 18) Write short answers to any NINE (9) questions:

(i) Define problem constraint.

(ii) Graph the solution set of the linear inequality 3y - 4 <= 0

(iii) Find slope of tangent to x2 + y2 = 5 at (4, 3)

(iv) Find alpha if u = alpha i + 2a j - k and v = i + alpha j + 3 k are perpendicular to each other. [sic, '2a' as printed]

(v) Find the direction cosine of the vector PQ, where P (2, 1, 5) and Q (1, 3, 1)

(vi) Find the vector from point A to origin where AB = 4i - 2j and B is the point (-2, 5)

(vii) Find cosine of the angle between u = [-3, 5] and v = [6, -2]

(viii) Write standard equation of the hyperbola.

(ix) Find the centre of the ellipse 9x2 + y2 = 18

(x) Find the equation of the circle with centre (5, -2) and radius is 4.

(xi) Find the equation of the hyperbola with foci (+-5, 0) and vertex (3, 0)

(xii) Find centre and radius of the circle 4x2 + 4y2 - 8x + 12y - 25 = 0

(xiii) Find focus and vertex of the parabola x2 = 5y

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (a) Prove that lim (x->0) (a^x - 1)/x = log_e a

(b) If x = (1-t2)/(1+t2), y = 2t/(1+t2), prove that y dy/dx + x = 0

Q6. (a) Evaluate integral of ln(x + sqrt(x2 + 1)) dx

(b) Prove that the linear equation ax + by + c = 0 in two variables x and y represents a straight line.

Q7. (a) Find the area between the x-axis and the curve y = sqrt(2ax - x2) when a > 0

(b) Graph the solution region of the system of linear inequalities and find the corner points of 2x - 3y <= 6, 2x + 3y <= 12, x >= 0

Q8. (a) Find a joint equation of the lines through the origin and perpendicular to the lines represented by x2 - 2xy tan alpha - y2 = 0

(b) Find equations of the tangent lines to the circle x2 + y2 - 4x + 2y = 0 [sign before 4x partially obscured] drawn from P (-1, 2)

Q9. (a) Find the centre, foci, eccentricity, vertices and equations of directrices of y2/16 - x2/9 = 1

(b) Prove that sin(alpha - beta) = sin alpha cos beta - cos alpha sin beta

GROUP – II

Objective Type

Time Allowed: 30 Min.	**Maximum Marks:** 20

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

> Note: Objective Type

Q1. (0,0) is the solution of inequality :

(A) 7x + 2y > 3
(B) x - 3y > 0
(C) x + 2y < 6
(D) x - 3y < 0

Q2. An equation of the horizontal line through the point P (7, -9) is :

(A) y = -9
(B) y = 9
(C) x = 7
(D) x = -7

Q3. Length of the vector 2i - j + 2k is:

(A) 6
(B) 4
(C) 3
(D) 5

Q4. Point of intersection of lines x - 2, y + 1 = 0 and 2x - y + 2 = 0 equals : [sic, as printed]

(A) (1, 0)
(B) (0, 1)
(C) (-1, 0)
(D) (0, -1)

Q5. Cosine of the angle between two non-zero vectors a and b is :

(A) a.b
(B) |a||b| / (a.b) [partially obscured in source scan]
(C) (a.b) / (|a||b|)
(D) (a x b) / (|a||b|)

Q6. The perpendicular distance of line 3x + 4 > + 10 = 0 from the origin is : [sic, as printed]

(A) 0
(B) 1
(C) 2
(D) 3

Q7. The radius of the circle x2 + y2 + 2gx + 2fy + c = 0 is :

(A) sqrt(g2 + f2 + c) [radical extent uncertain in source scan]
(B) sqrt(g2 - f2 + c)
(C) g + f - c
(D) sqrt(g2 + f2 - c)

Q8. The condition for a line y = mx + c to be the tangent to the circle x2 + y2 = a2 is :

(A) c +- m sqrt(1+a2) [watermarked, uncertain]
(B) c +- a sqrt(1+m2)
(C) c = +-a sqrt(1-m2)
(D) c = +-m sqrt(1-a2)

Q9. Slope of line perpendicular to line 3x - 4y + 5 = 0 is :

(A) -3/4
(B) -4/3
(C) 3/4
(D) 4/3

Q10. In an ellipse, the foci lie on:

(A) Major axis
(B) Minor axis
(C) Directrix
(D) Z-axis

Q11. d/dx [f(x)/g(x)] = :

(A) (f(x)g'(x) - f'(x)g(x)) / [g(x)]2
(B) (f'(x)g(x) - f(x)g'(x)) / [f(x)]2
(C) (g(x)f'(x) - f(x)g'(x)) / [g(x)]2
(D) (g'(x)f(x) - g(x)f(x)) / [g(x)]2

Q12. If the degree of a polynomial function is 1, then it is called :

(A) Identity function
(B) Linear function
(C) Constant function
(D) Trigonometric function

Q13. Integral of sin 3x dx = :

(A) -cos3x/3 + c
(B) cos3x/3 + c
(C) 3 cos 3x + c
(D) -3 cos 3x + c

Q14. If y = 1/x2, then dy/dx at x = 1 is:

(A) 2 [as printed; sign obscured]
(B) 3
(C) 1/3
(D) 4

Q15. The solution of differential equation dy/dx = -y is:

(A) y = xe^-x
(B) y = ce^-x
(C) y = e^x
(D) y = ce^x

Q16. Two positive integer whose sum is 30 and their product will be maximum are :

(A) 14, 16
(B) 15, 15
(C) 10, 20
(D) 12, 18

Q17. Integral from -1 to 3 of x3 dx = :

(A) 20
(B) 40
(C) 30
(D) 80

Q18. lim (x->1) (x2 - 1)/(x2 - x) = :

(A) 2
(B) 1/2
(C) 4
(D) 5

Q19. Integral of sec x dx = :

(A) ln(sec x + tan x) + c
(B) ln(cosec x + cot x) + c
(C) ln(sin x + cos x) + c
(D) sec x + tan x + c

Q20. d/dx (cot-1 x) = :

(A) 1/(1+x2)

(B) -1/(1+x2)

(C) -cosec2x

(D) sec2x

Essay Type

Time Allowed: 2.30 hrs. | **Maximum Marks:** 80

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Find domain and range of f(x) = sqrt(x + 1)

(ii) Find fof(x) if f(x) = sqrt(x + 1)

(iii) Obtain f-1(x) from f(x) = 3x3 + 7

(iv) Evaluate lim (theta->0) (1 - cos theta)/theta

(v) Express lim (x->infinity) (x/(1+x))^x in terms of "e"

(vi) If y = (x2 + 1)/(x2 - 3), then find dy/dx

(vii) Prove that derivative of tan-1x w.r.t. "x" is 1/(1+x2)

(viii) Differentiate sin-1(a/x) w.r.t. "x" [sic, stray mark before sin-1 in source scan]

(ix) Find dy/dx if y = x2 ln sqrt(x)

(x) If y = e^-x (x3 + 2x2 + 1), then find dy/dx

(xi) Apply the Maclaurin's series expansion to prove that e^x = 1 + x/1! + x2/2! + x3/3i + ----- [sic, '3i' as printed]

(xii) Determine the interval in which f(x) = sinx, x in (-pi, pi) is decreasing.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) If x2 + 2y2 = 16, find dy/dx by using differentials.

(ii) Evaluate integral of x/(x + 2) dx

(iii) Evaluate indefinite integral of sec2x / sqrt(tanx) dx

(iv) Evaluate integral of ln x dx

(v) Evaluate the definite integral from -1 to 1 of (x^(1/3) + 1) dx

(vi) Find the area between the x-axis and the curve y = x2 + 1 from x = 1 to x = 2

(vii) Evaluate integral of e^-x (cos x - sin x) dx

(viii) Solve x dy + y(x - 1)dx = 0

(ix) Show that the points A(3, 1), B(-2, -3) and C(2, 2) are vertices of an isosceles triangle.

(x) Find an equation of line having x-intercept: -9 and slope: -4

(xi) Show that the lines 4x - 3y - 8 = 0, 3x - 4y - 6 = 0 and x - y - 2 = 0 are concurrent.

(xii) What is homogeneous equation?

Q4. (Marks 18) Write short answers to any NINE (9) questions:

(i) Graph the solution set of 2x + 1 >= 0

(ii) Define problem constraint.

(iii) Find an equation of circle with centre (sqrt(2), -3 sqrt(3)) and radius 2 sqrt(2)

(iv) Find slope of tangent to x2 + y2 = 5 at (4, 3)

(v) Check the position of the point (5, 6) with respect to the circle x2 + y2 = 81

(vi) Find focus and vertex of y2 = 8x

(vii) Find equation of ellipse with foci (+-3, 0) and minor axis of length 10.

(viii) Find equation of hyperbola with centre (0, 0), focus (6, 0), vertex (4, 0)

(ix) Find a vector from the point A to the origin where AB vector = 4i - 2j and B (-2, 5)

(x) Find alpha so that |alpha i + (alpha + 1) j + 2 k| = 3

(xi) Find the cosine of the angle theta between u and v; u = i - 3j + 4k; v = 4i - j + 3k

(xii) Prove that a.(b + c) + b x (c + a) + c x (a + b) = 0 [sic, first operator partially obscured]

(xiii) A force F = 7i + 4j - 3k is applied at P (1, -2, 3). Find its moment about the point Q (2, 1, 1)

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (a) Discuss the continuity of f(x) at x = 1: f(x) = {3x - 1 if x < 1; 4 if x = 1; 4x if x > 1}

(b) Show that 2^(x+h) = 2^x {1 + (ln2)h + (ln2)2 h2/2! + (ln2)3 h3/3! + ...}

Q6. (a) Evaluate integral of sqrt(4 - 5x2) dx

(b) Find the equation of perpendicular bisector of segment joining the points A (3, 5) and B (9, 8)

Q7. (a) Evaluate the integral from 0 to pi/4 of (cos theta + sin theta)/(2 cos2 theta) d theta

(b) Maximize f(x, y) = x + 3y subject to the constraints 2x + 5y <= 30; 5x + 4y <= 20, x >= 0, y >= 0

Q8. (a) Find the interior angles whose vertices are A(-2, 11), B(-6, -3), C(4, -9)

(b) Find an equation of the circle passing through the points A (4, 5), B (-4, -3), C (8, -3)

Q9. (a) Prove angle in a semi circle is right angle.

(b) Find an equation of the tangent to the parabola y2 = -6x which is parallel to the line 2x + y + 1 = 0. Also find point of tangency.